

- 6 (a) 6 men take 4 hours to dig a trench 6 m deep. How long will it take 10 men to dig a trench 5 m deep?

*Answer* ..... hours [2]

- (b) It is given that  $y$  varies directly as the square root of  $x$ . Given that  $y = 24$  for a particular value of  $x$ . Find the value of  $y$  when  $x$  is increased by 300%.

*Answer*  $y =$  ..... [2]

- 7 A map has a scale of 1: 30 000. The length of a rectangular park is 8 cm on the map.

- (a) Find the actual length of the park in km.

*Answer* ..... km [1]

- (b) Given that the park covers an area of  $4.5 \text{ km}^2$ . Find the breadth of the park on the map.

*Answer* ..... cm [2]

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- 8 (a) Convert 9 km/h into m/s.

*Answer* ..... m/s [1]

- (b) A car uses 17.25 litres of fuel to travel a distance of 250 km. Calculate the distance the car can travel with 60 litres of fuel.

*Answer* ..... km [1]

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- 9 Express as a single fraction in its simplest form  $\frac{1}{18x^2 - 2} + \frac{2}{3x - 1}$ .

*Answer* ..... [3]

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- 10 (a) Factorise  $4a^2b + 4a^2 - b - 1$  completely.

*Answer* ..... [2]

- (b) Given that  $f = 3g - 2x^3$ , express  $x$  in terms of  $f$  and  $g$ .

*Answer* ..... [2]

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